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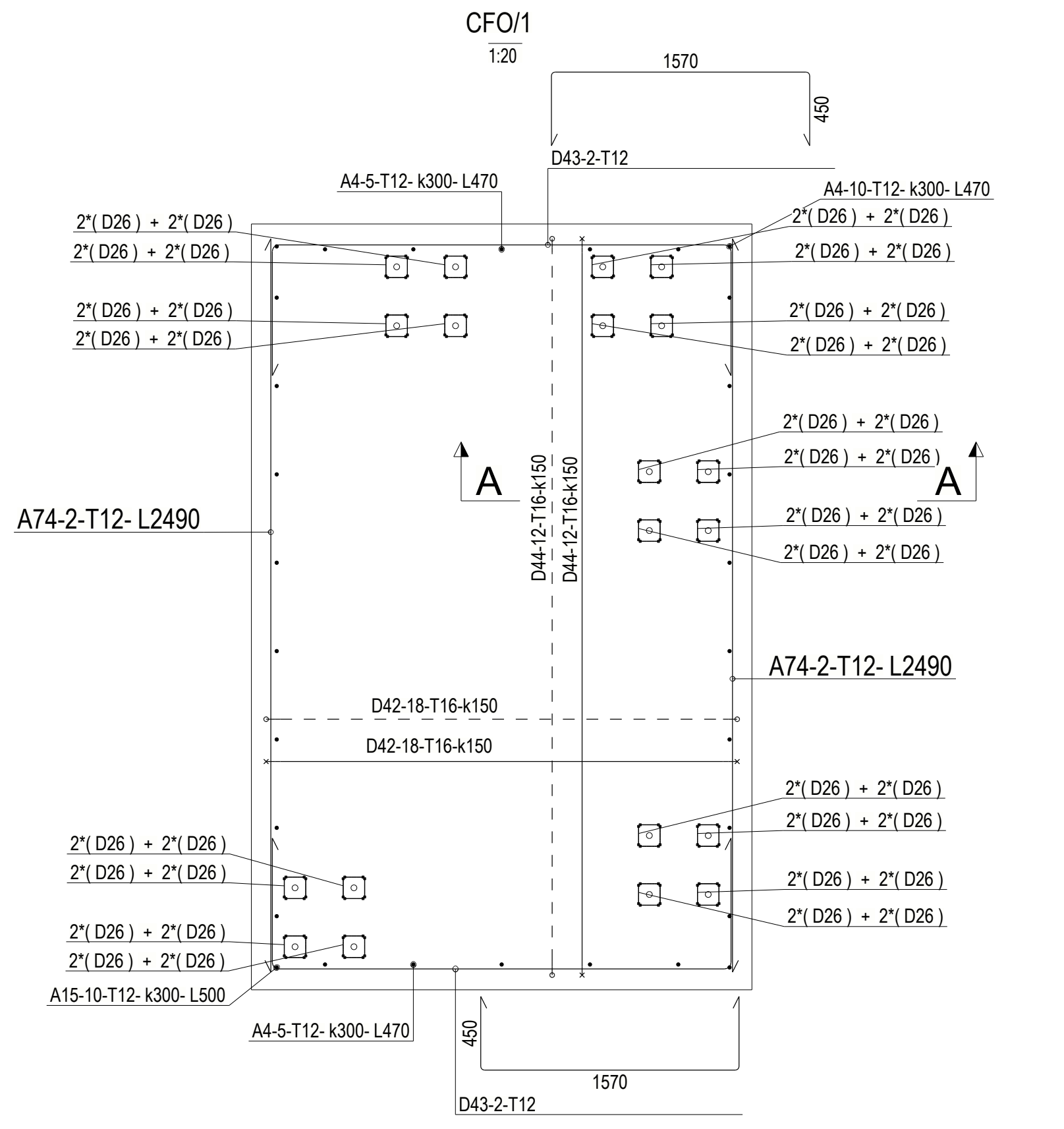
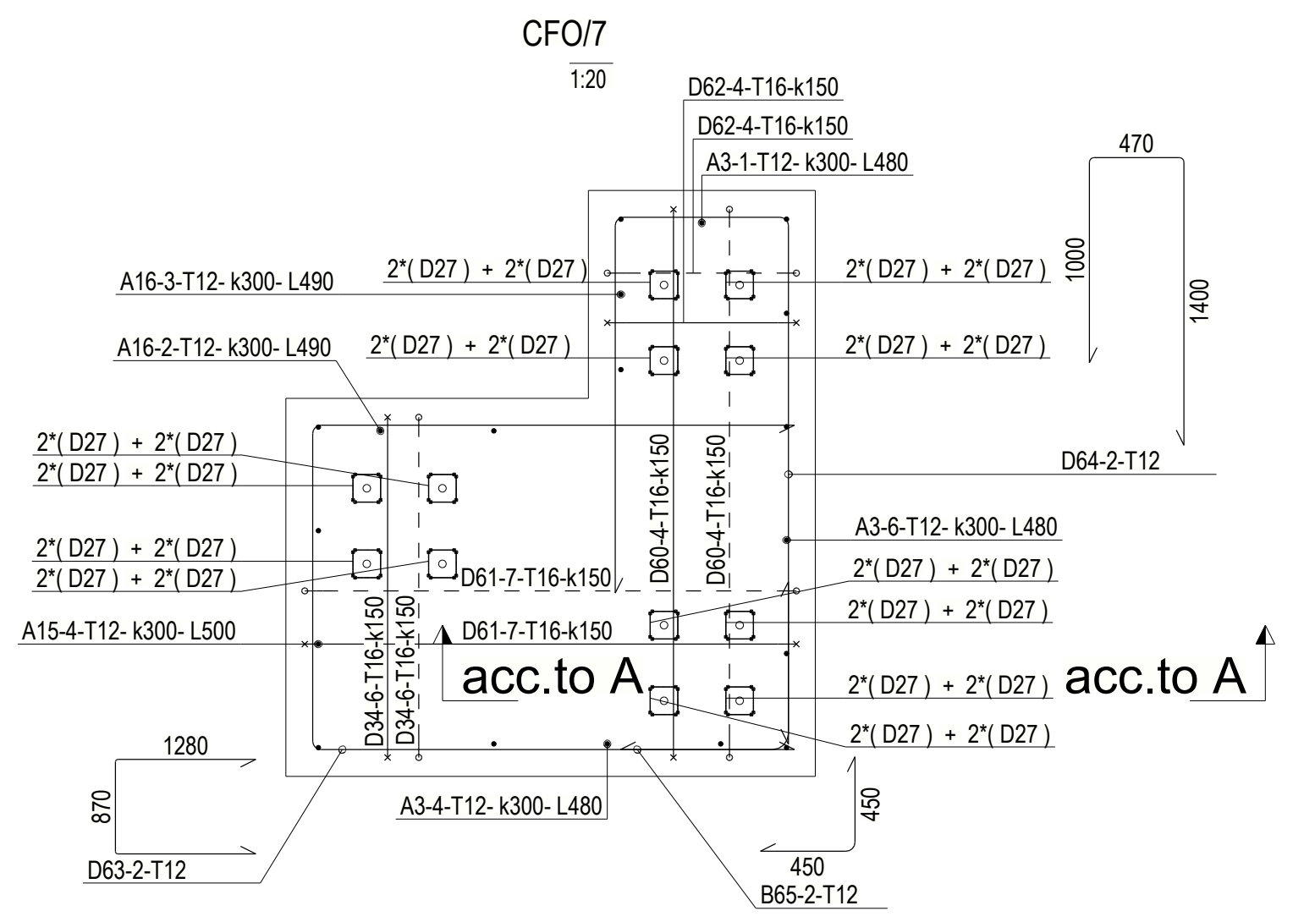
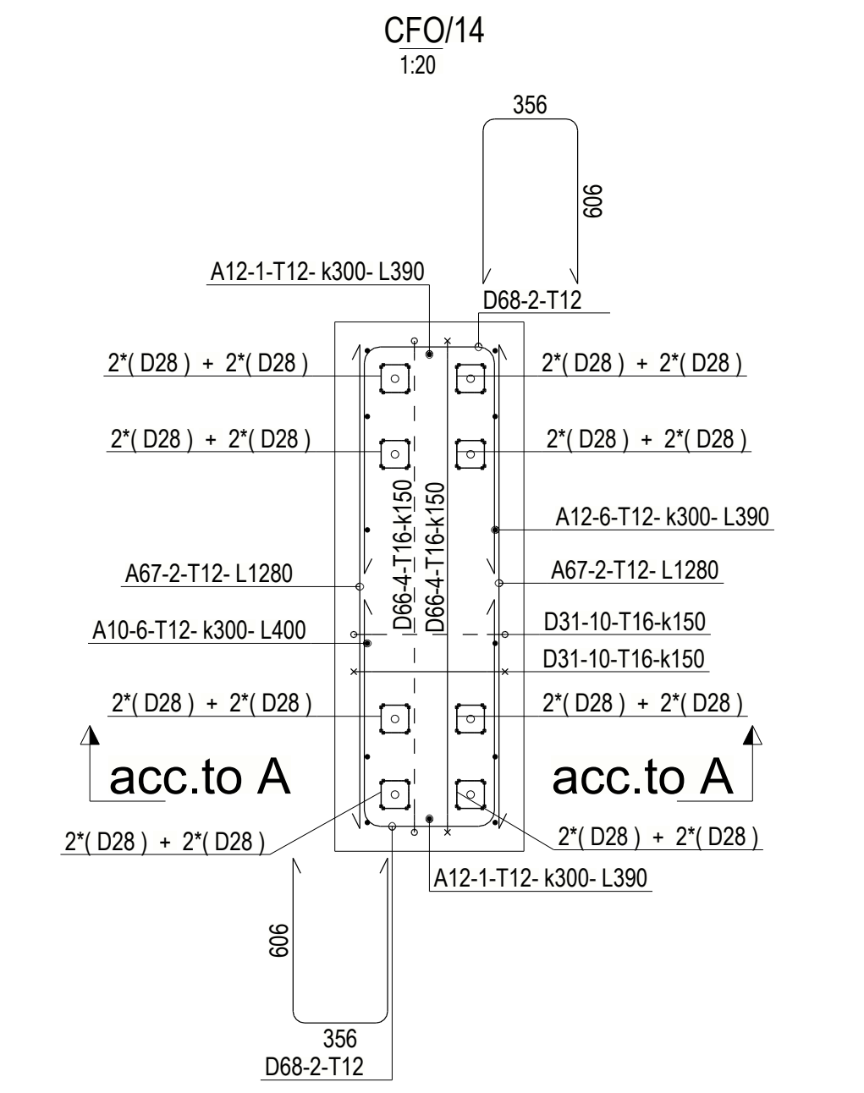
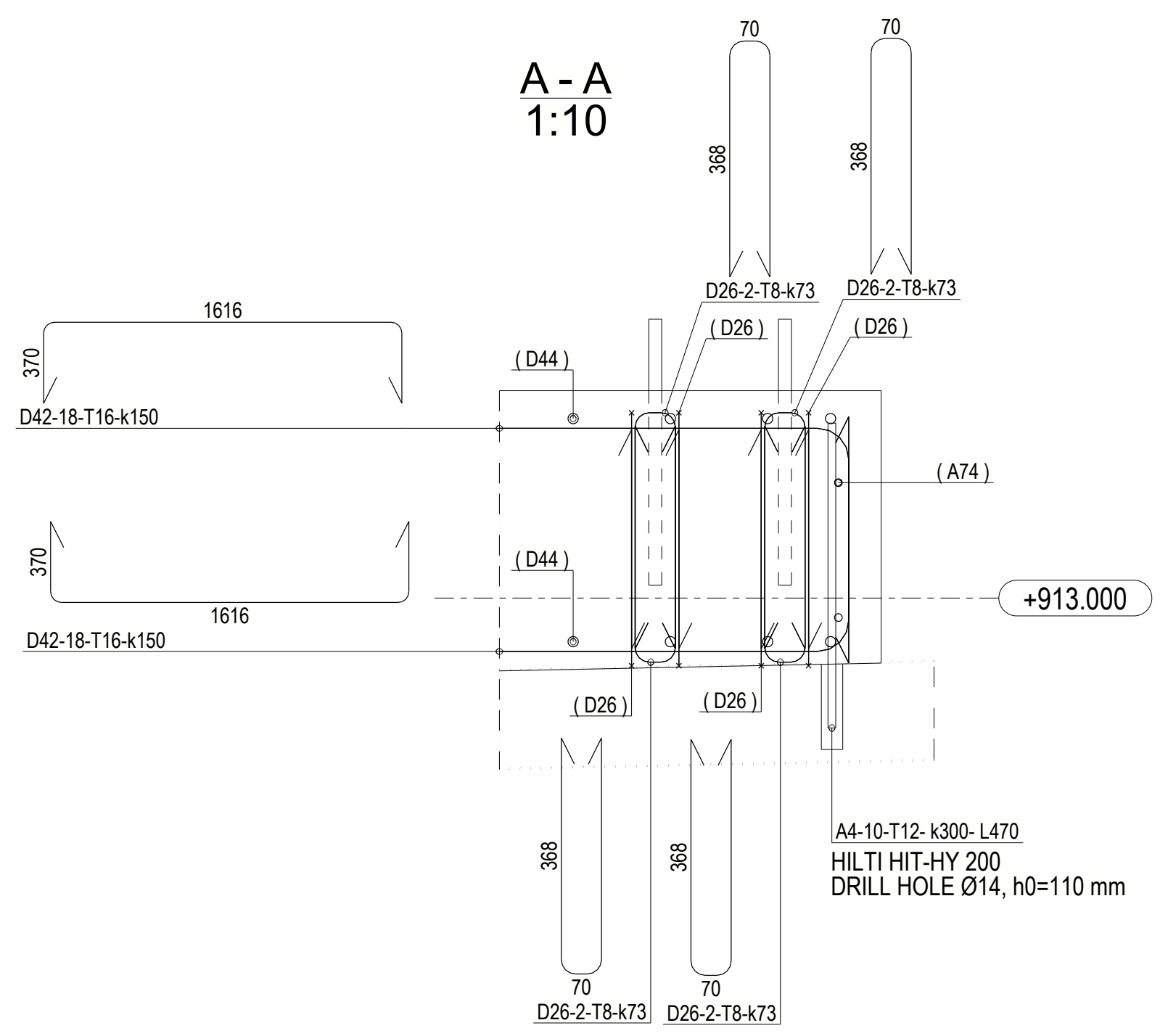
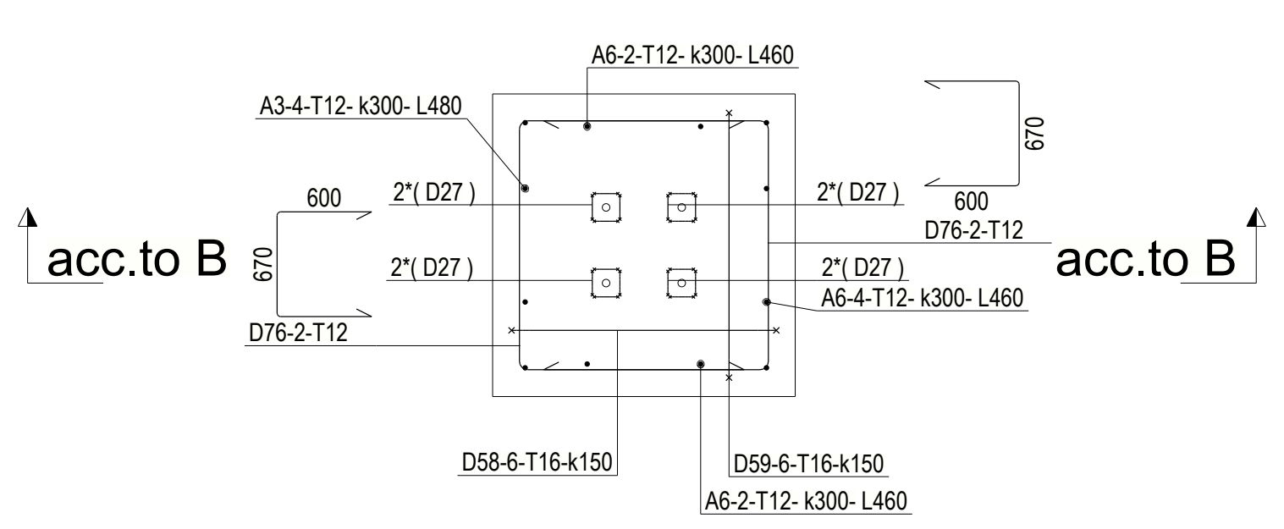
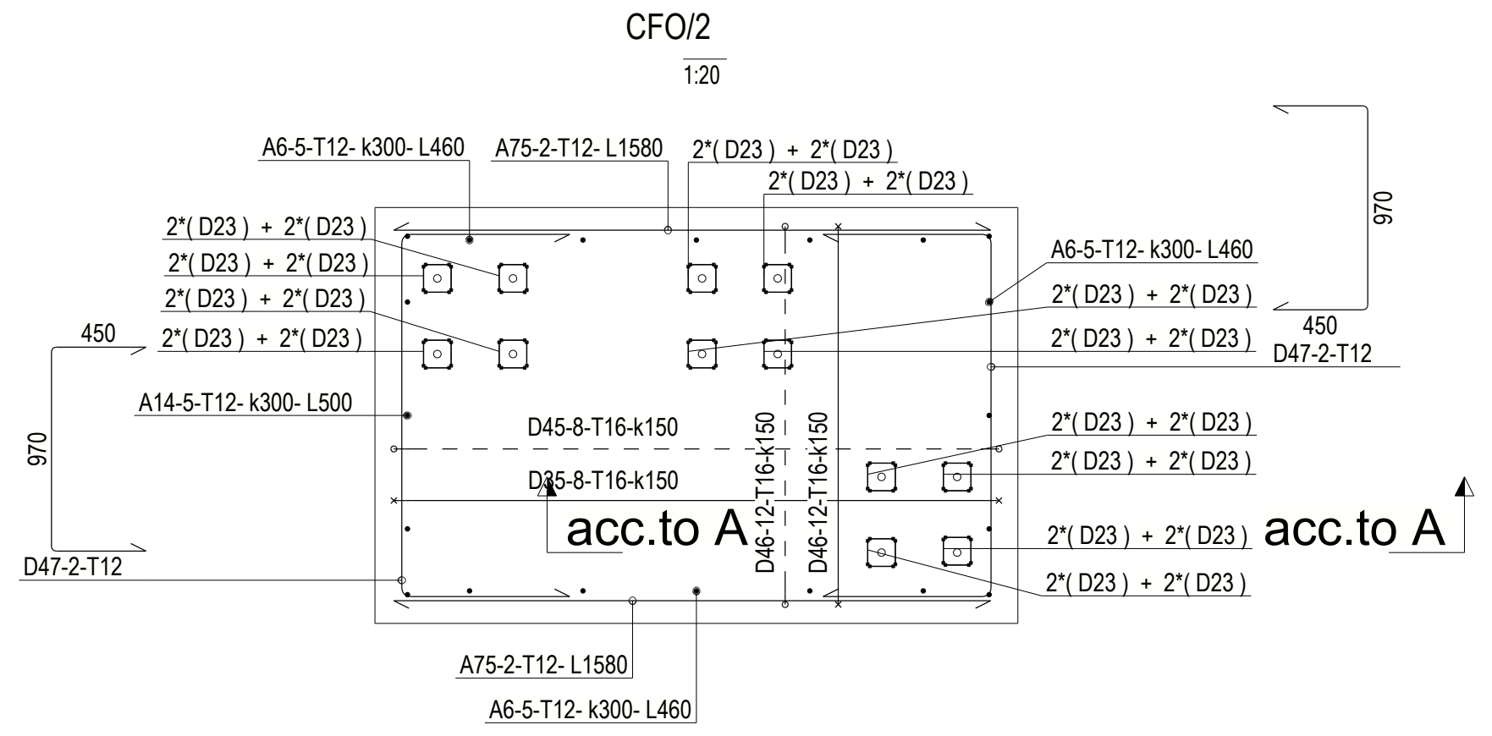


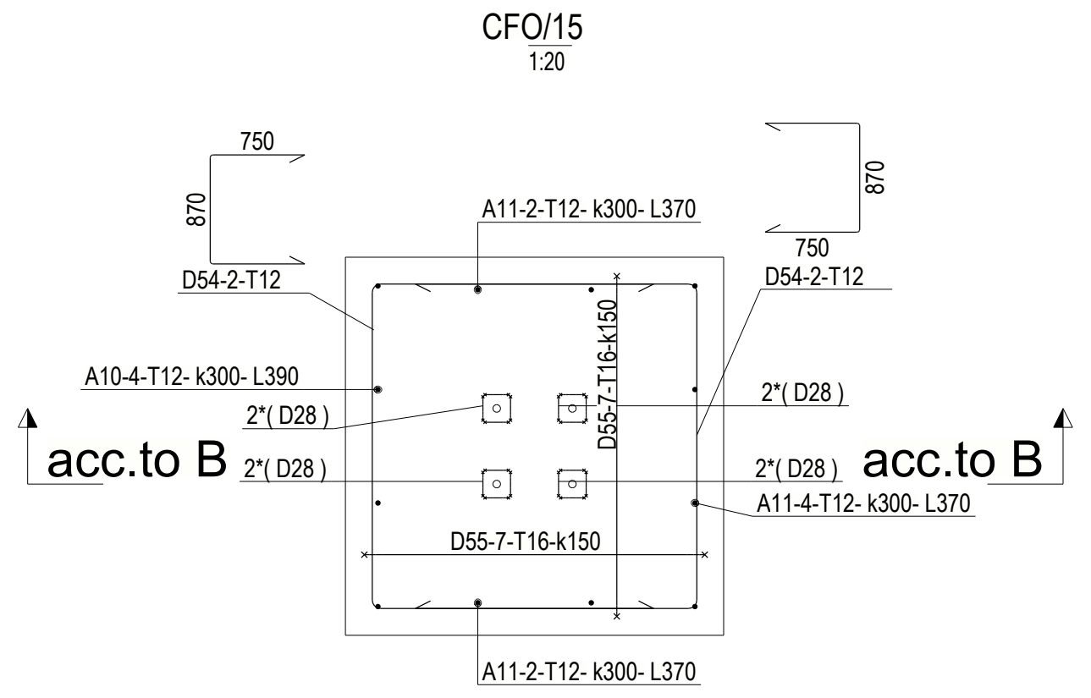


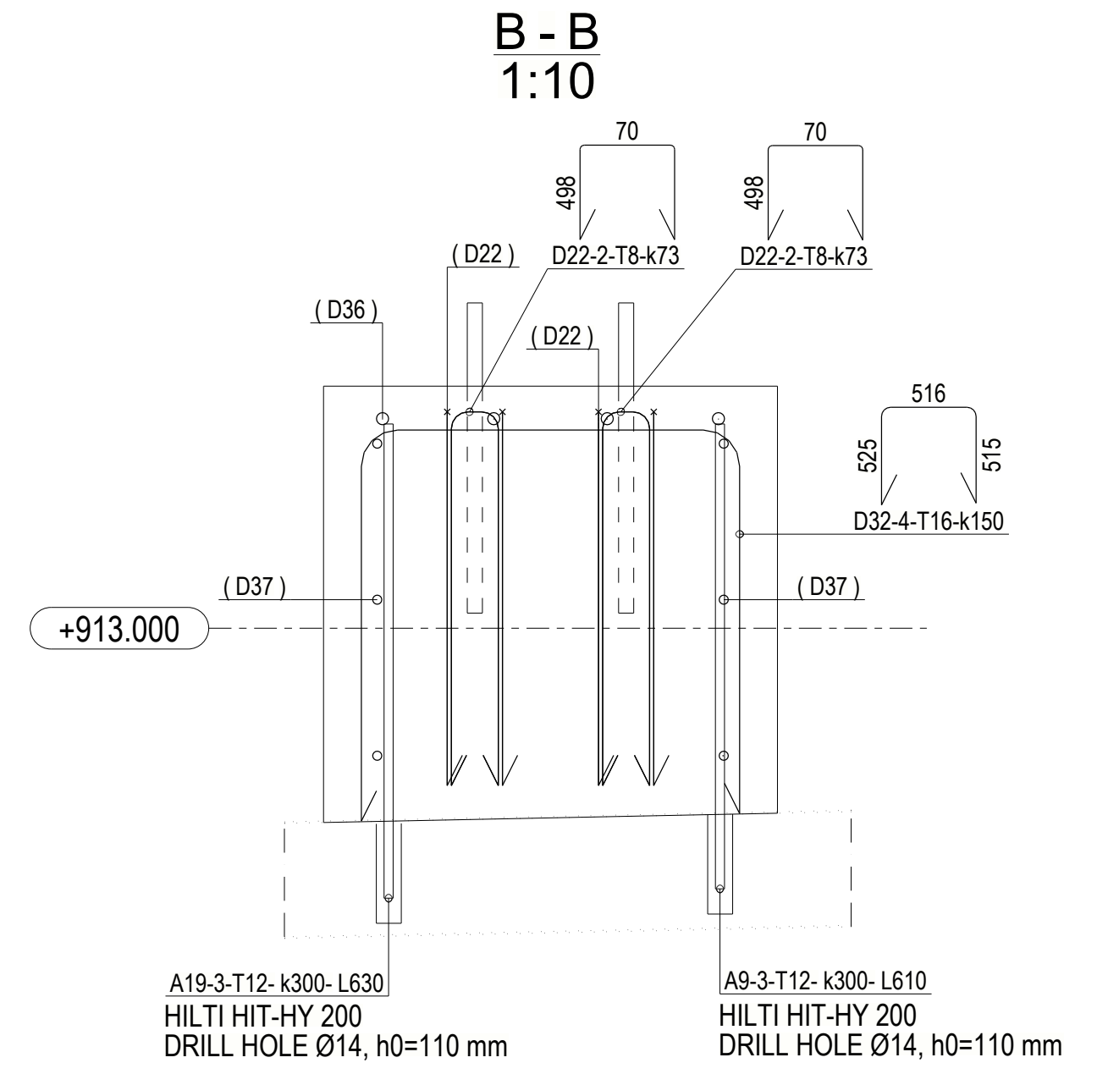


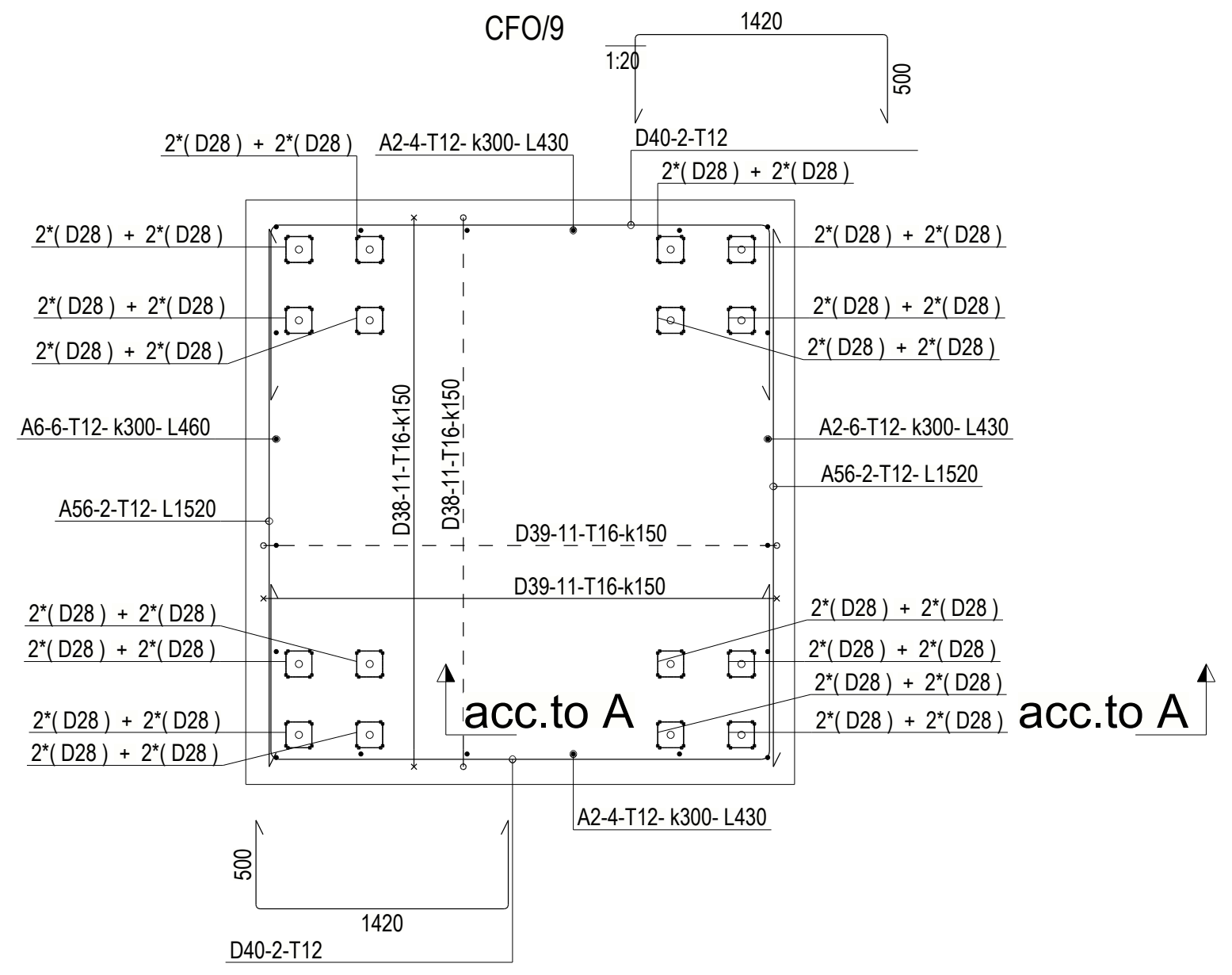


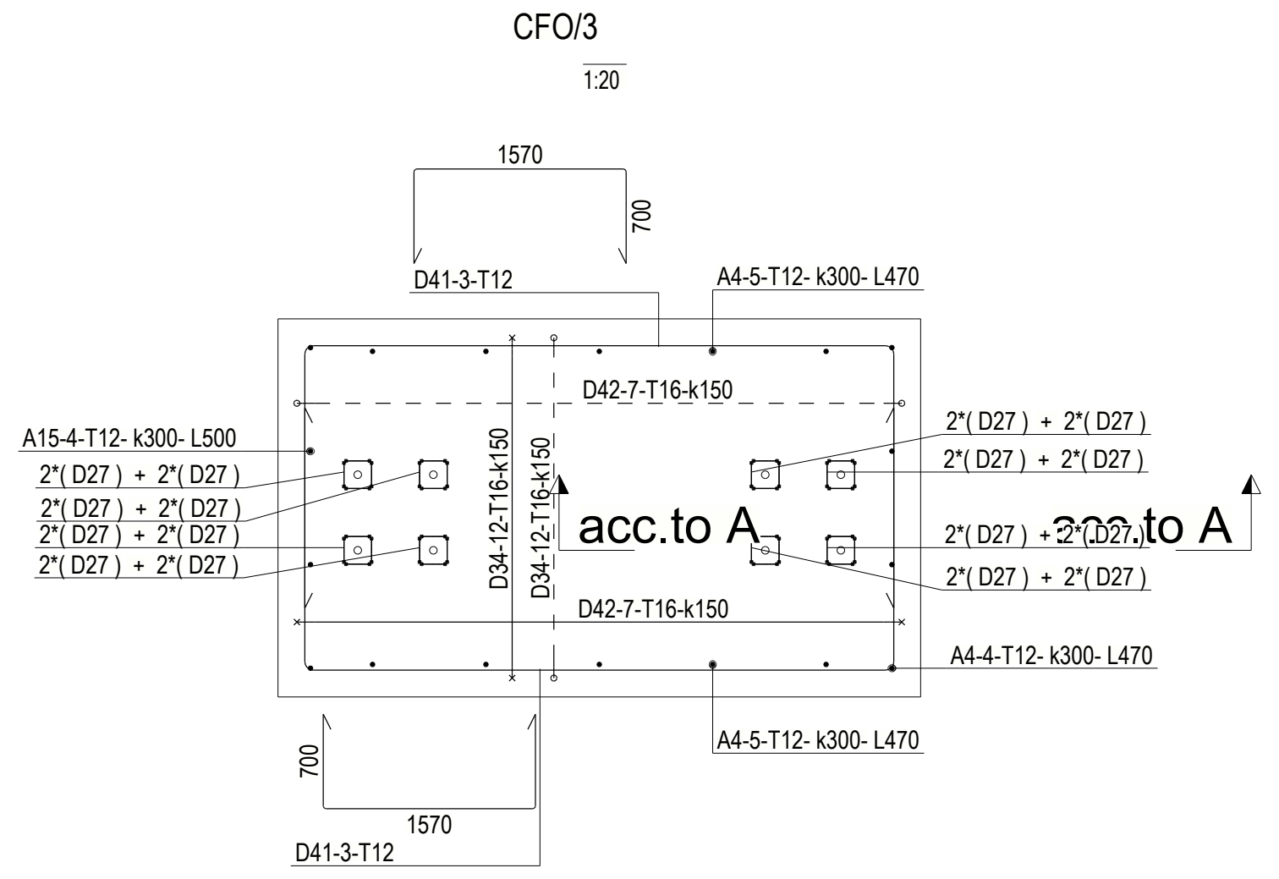


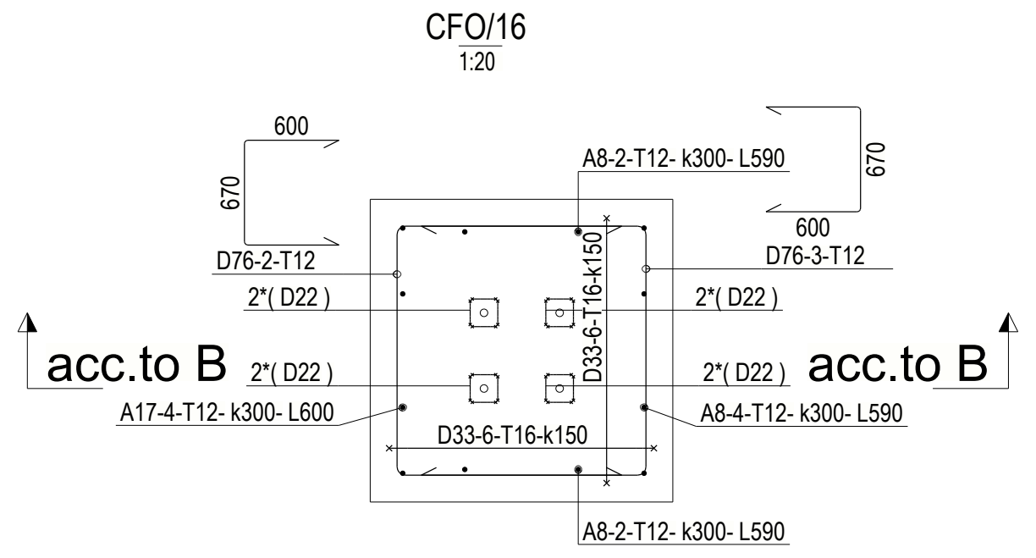


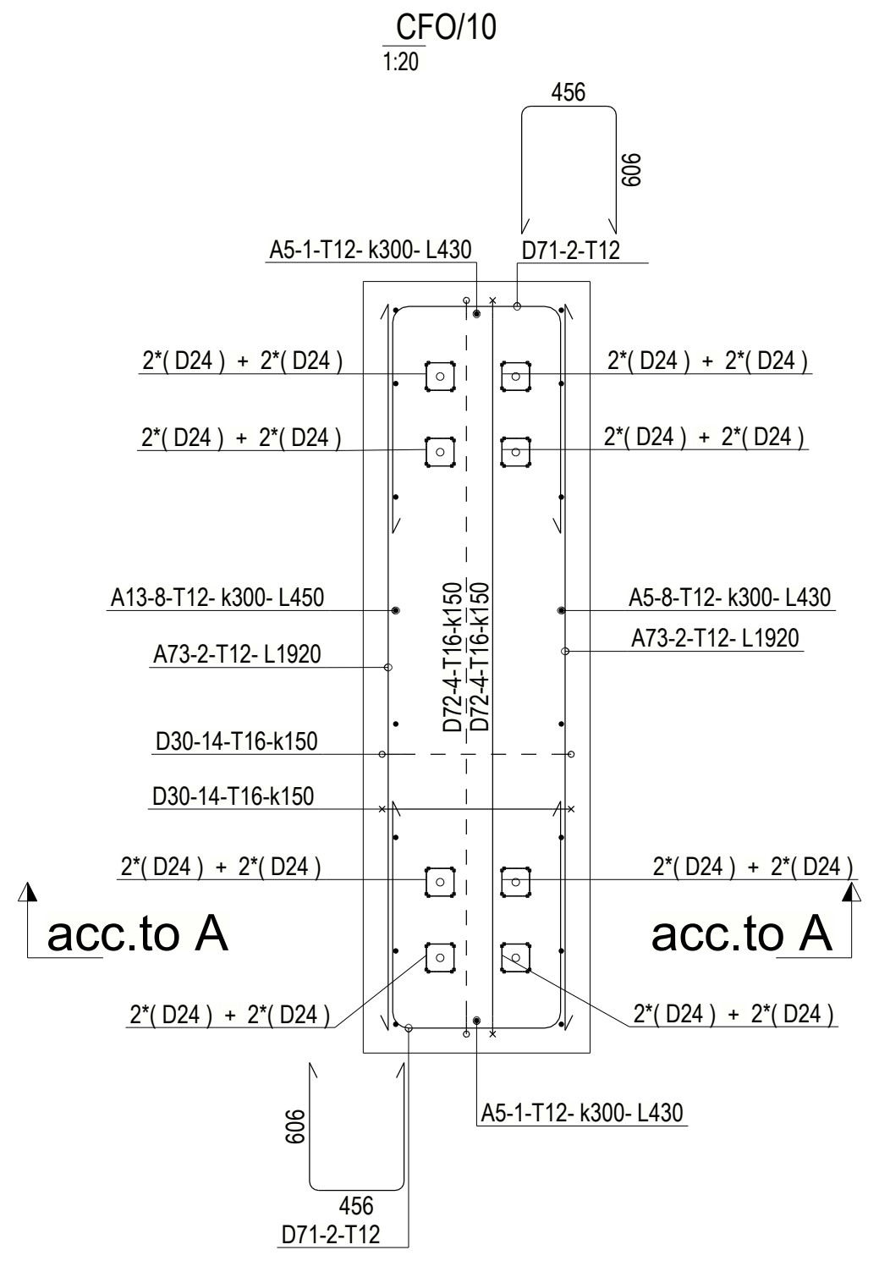


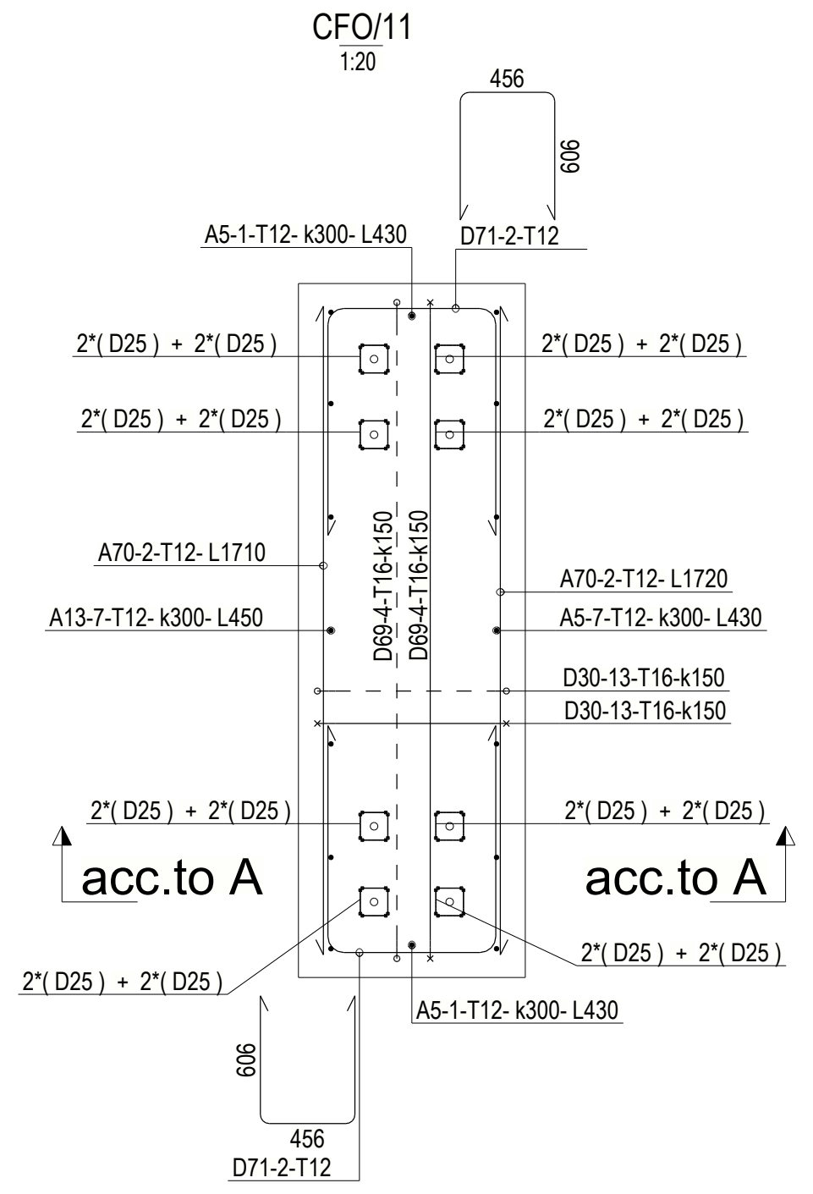


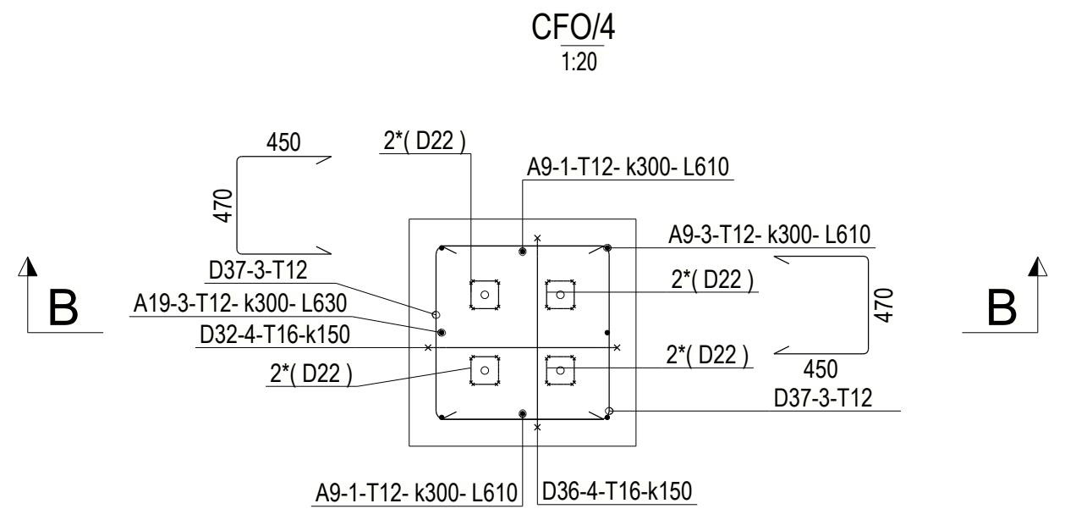


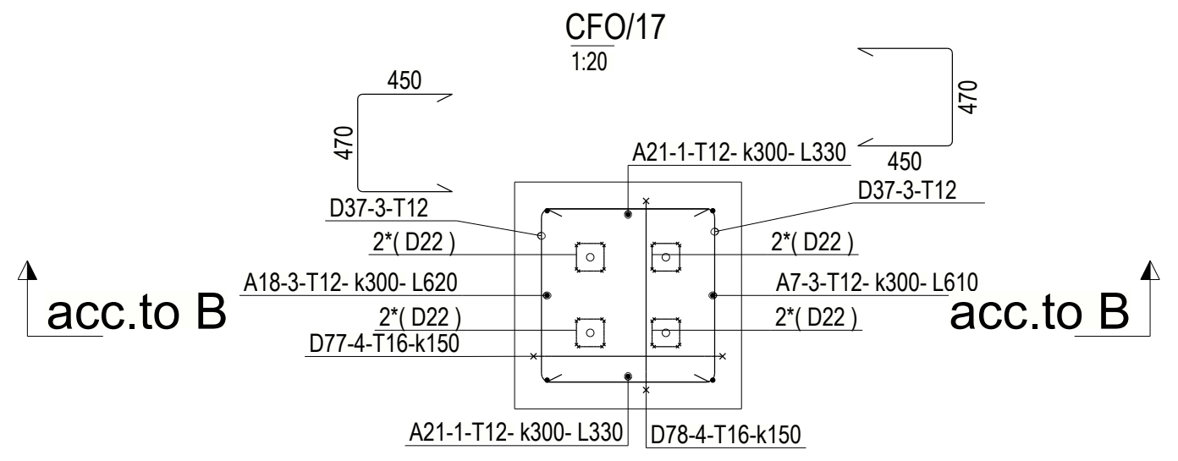


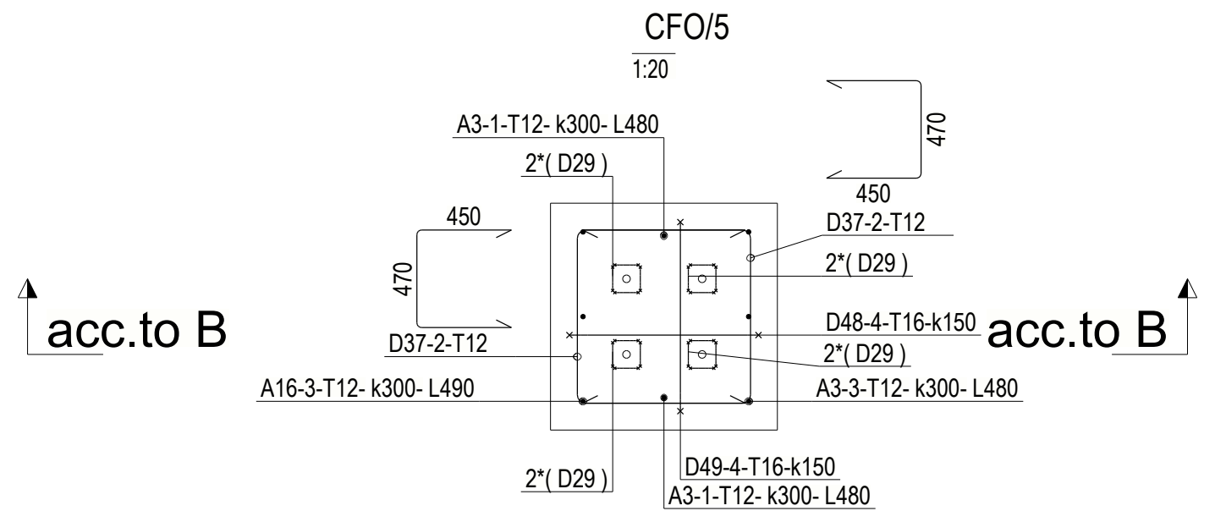


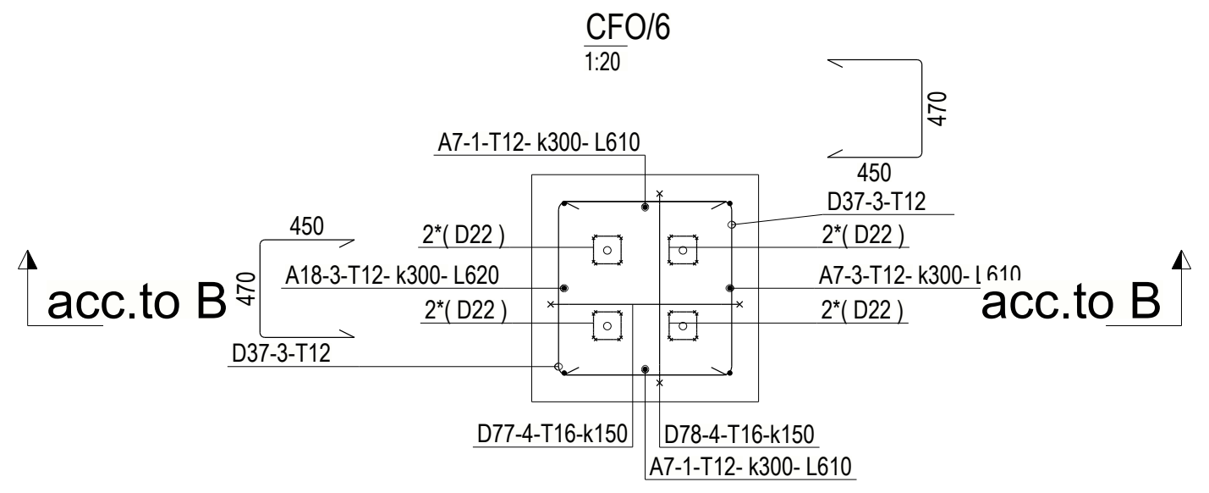


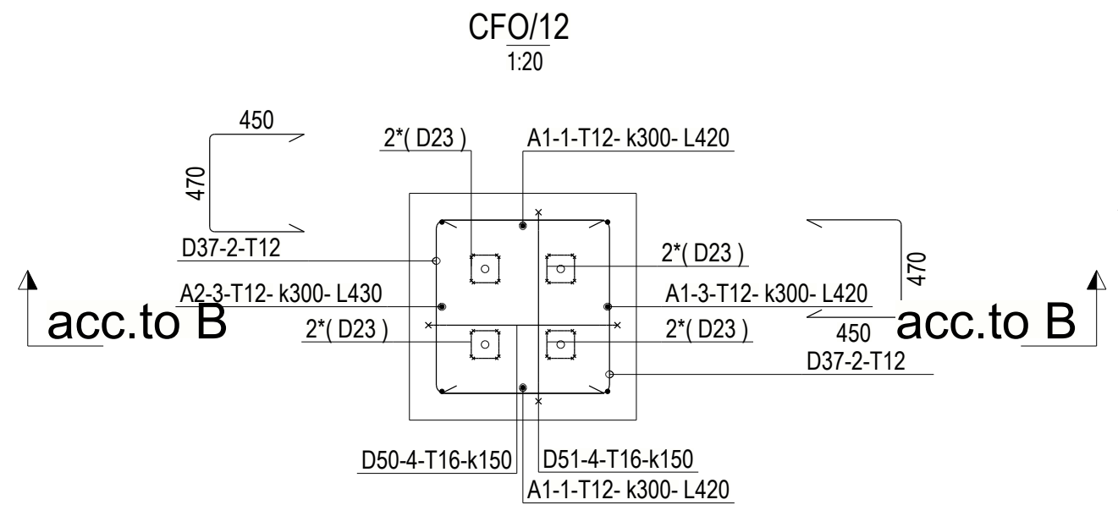


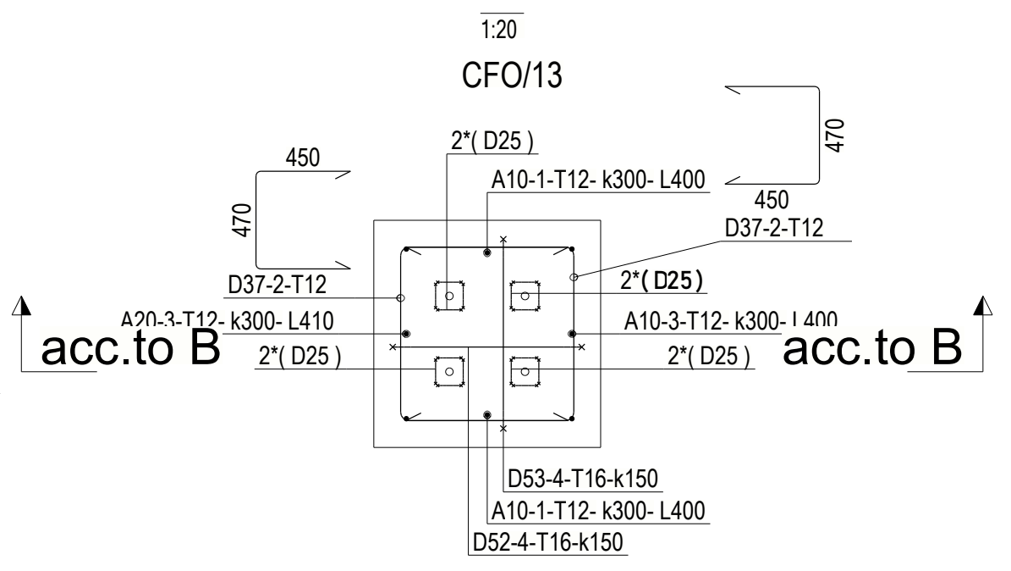












REINFORCING BAR LIST CAST UNIT CODE: CFO/2 PCS: 1

REINFORCING BARS				D	L	dL	WEIGHT	BENDING DIMENSIONS [mm]									COMMENT
TYPE	POS	PCS	GRADE	[mm]	[mm]	[mm]	SUM [kg]	a	b	c	d	e	u	v	x	TD	
A	6	15	AISI304	12	460		6.1	461									
A	14	5	AISI304	12	500		2.2	496									
D	23	96	AISI304	8	730		27.7	348	70	348						36	
D	35	8	AISI304	16	2190		27.7	326	1616	326						80	
D	45	8	AISI304	16	2260		28.6	360	1616	360						80	
D	46	24	AISI304	16	1660		63.1	360	1016	360						80	
D	47	4	AISI304	12	1830		6.5	450	970	450						32	
A	75	4	AISI304	12	1580		5.6	1578									

RAUDOITTEIDEN KOKONAISPAINO YHTEENSÄ [kg]: 167.6

REINFORCING BAR LIST CAST UNIT CODE: CFO/3 PCS: 1

REINFORCING BARS				D	L	dL	WEIGHT	BENDING DIMENSIONS [mm]									COMMENT
TYPE	POS	PCS	GRADE	[mm]	[mm]	[mm]	SUM [kg]	a	b	c	d	e	u	v	x	TD	
A	4	14	AISI304	12	470		5.8	469									
A	15	4	AISI304	12	500		1.8	504									
D	27	64	AISI304	8	790		20.0	378	70	378						36	
D	34	24	AISI304	16	1490		56.7	326	916	326						80	
D	41	6	AISI304	12	2930		15.6	700	1570	700						32	
D	42	14	AISI304	16	2280		50.5	370	1616	370						80	

RAUDOITTEIDEN KOKONAISPAINO YHTEENSÄ [kg]: 150.4

REINFORCING BAR LIST CAST UNIT CODE: CFO/4 PCS: 1

REINFORCING BARS				D	L	dL	WEIGHT	BENDING DIMENSIONS [mm]									COMMENT
TYPE	POS	PCS	GRADE	[mm]	[mm]	[mm]	SUM [kg]	a	b	c	d	e	u	v	x	TD	
A	9	5	AISI304	12	610		2.7	615									
A	19	3	AISI304	12	630		1.7	627									
D	22	16	AISI304	8	1030		6.5	498	70	498						36	
D	32	4	AISI304	16	1480		9.4	525	516	515						80	
D	36	4	AISI304	16	1500		9.5	530	516	530						80	
D	37	6	AISI304	12	1330		7.1	450	470	450						32	

RAUDOITTEIDEN KOKONAISPAINO YHTEENSÄ [kg]: 36.9

REINFORCING BAR LIST CAST UNIT CODE: CFO/5 PCS: 1

REINFORCING BARS				D	L	dL	WEIGHT	BENDING DIMENSIONS [mm]									COMMENT
TYPE	POS	PCS	GRADE	[mm]	[mm]	[mm]	SUM [kg]	a	b	c	d	e	u	v	x	TD	
A	3	5	AISI304	12	480		2.1	477									
A	16	3	AISI304	12	490		1.3	490									
D	29	16	AISI304	8	830		5.3	398	70	398						36	
D	37	4	AISI304	12	1330		4.7	450	470	450						32	
D	48	4	AISI304	16	1200		7.6	385	516	375						80	
D	49	4	AISI304	16	1230		7.8	395	516	395						80	

RAUDOITTEIDEN KOKONAISPAINO YHTEENSÄ [kg]: 28.8

REINFORCING BAR LIST CAST UNIT CODE: CFO/6 PCS: 1

REINFORCING BARS				D	L	dL	WEIGHT	BENDING DIMENSIONS [mm]									COMMENT
TYPE	POS	PCS	GRADE	[mm]	[mm]	[mm]	SUM [kg]	a	b	c	d	e	u	v	x	TD	
A	7	5	AISI304	12	610		2.7	609									
A	18	3	AISI304	12	620		1.7	621									
D	22	16	AISI304	8	1040		6.5	500	70	500						36	
D	37	6	AISI304	12	1330		7.1	450	470	450						32	
D	77	4	AISI304	16	1460		9.2	510	516	510						80	
D	78	4	AISI304	16	1480		9.4	520	516	520						80	

RAUDOITTEIDEN KOKONAISPAINO YHTEENSÄ [kg]: 36.6

REINFORCING BAR LIST CAST UNIT CODE: CFO/7 PCS: 1

REINFORCING BARS				D	L	dL	WEIGHT	BENDING DIMENSIONS [mm]									COMMENT
TYPE	POS	PCS	GRADE	[mm]	[mm]	[mm]	SUM [kg]	a	b	c	d	e	u	v	x	TD	
A	3	11	AISI304	12	480		4.6	476									
A	15	4	AISI304	12	500		1.8	504									
A	16	5	AISI304	12	490		2.2	488									
D	27	96	AISI304	8	790		30.0	378	70	378						36	
D	34	12	AISI304	16	1490		28.3	326	916	326						80	
D	60	8	AISI304	16	2040		25.8	326	1466	326						80	
D	61	14	AISI304	16	1940		43.0	350	1316	350						80	
D	62	8	AISI304	16	1140		14.4	350	516	350						80	
D	63	2	AISI304	12	3390		6.0	1280	870	1280						32	
D	64	2	AISI304	12	2830		5.0	1400	470	1000						32	
B	65	2	AISI304	12	870		1.5	450	450							72	

RAUDOITTEIDEN KOKONAISPAINO YHTEENSÄ [kg]: 162.8

REINFORCING BAR LIST CAST UNIT CODE: CFO/8 PCS: 1

REINFORCING BARS				D	L	dL	WEIGHT	BENDING DIMENSIONS [mm]									COMMENT
TYPE	POS	PCS	GRADE	[mm]	[mm]	[mm]	SUM [kg]	a	b	c	d	e	u	v	x	TD	
A	3	4	AISI304	12	480		1.7	479									
A	6	8	AISI304	12	460		3.3	462									
D	27	16	AISI304	8	790		5.0	378	70	378						36	
D	58	6	AISI304	16	1390		13.2	380	716	370						80	
D	59	6	AISI304	16	1420		13.5	391	716	391						80	
D	76	4	AISI304	12	1830		6.5	600	670	600						32	

RAUDOITTEIDEN KOKONAISPAINO YHTEENSÄ [kg]: 43.2

REINFORCING BAR LIST CAST UNIT CODE: CFO/9 PCS: 1

REINFORCING BARS				D	L	dL	WEIGHT	BENDING DIMENSIONS [mm]									COMMENT
TYPE	POS	PCS	GRADE	[mm]	[mm]	[mm]	SUM [kg]	a	b	c	d	e	u	v	x	TD	
A	2	14	AISI304	12	430		5.3	429									
A	6	6	AISI304	12	460		2.5	460									
D	28	128	AISI304	8	630		31.9	298	70	298						36	
D	38	22	AISI304	16	2190		76.2	350	1566	350						80	
D	39	22	AISI304	16	2090		72.7	350	1466	350						80	
D	40	4	AISI304	12	2380		8.4	500	1420	500						32	
A	56	4	AISI304	12	1520		5.4	1518									

RAUDOITTEIDEN KOKONAISPAINO YHTEENSÄ [kg]: 202.5

REINFORCING BAR LIST CAST UNIT CODE: CFO/13 PCS: 1

REINFORCING BARS				D	L	dL	WEIGHT	BENDING DIMENSIONS [mm]									COMMENT
TYPE	POS	PCS	GRADE	[mm]	[mm]	[mm]	SUM [kg]	a	b	c	d	e	u	v	x	TD	
A	10	5	AISI304	12	400		1.8	399									
A	20	3	AISI304	12	410		1.1	411									
D	25	16	AISI304	8	690		4.4	328	70	328						36	
D	37	4	AISI304	12	1330		4.7	450	470	450						32	
D	52	4	AISI304	16	1050		6.7	310	516	300						80	
D	53	4	AISI304	16	1070		6.8	315	516	315						80	

RAUDOITTEIDEN KOKONAISPAINO YHTEENSÄ [kg]: 25.4

REINFORCING BAR LIST CAST UNIT CODE: CFO/14 PCS: 1

REINFORCING BARS				D	L	dL	WEIGHT	BENDING DIMENSIONS [mm]									COMMENT
TYPE	POS	PCS	GRADE	[mm]	[mm]	[mm]	SUM [kg]	a	b	c	d	e	u	v	x	TD	
A	10	6	AISI304	12	400		2.1	400									
A	12	8	AISI304	12	390		2.8	393									
D	28	64	AISI304	8	630		16.0	298	70	298						36	
D	31	20	AISI304	16	940		29.8	300	416	300						80	
D	66	8	AISI304	16	1840		23.3	300	1316	300						80	
A	67	4	AISI304	12	1280		4.5	1280									
D	68	4	AISI304	12	1510		5.4	606	356	606						72	

RAUDOITTEIDEN KOKONAISPAINO YHTEENSÄ [kg]: 83.9

REINFORCING BAR LIST CAST UNIT CODE: CFO/15 PCS: 1

REINFORCING BARS				D	L	dL	WEIGHT	BENDING DIMENSIONS [mm]									COMMENT
TYPE	POS	PCS	GRADE	[mm]	[mm]	[mm]	SUM [kg]	a	b	c	d	e	u	v	x	TD	
A	10	4	AISI304	12	390		1.4	394									
A	11	8	AISI304	12	370		2.7	374									
D	28	16	AISI304	8	630		4.0	298	70	298						36	
D	54	4	AISI304	12	2330		8.3	750	870	750						32	
D	55	14	AISI304	16	1410		31.4	285	916	285						80	

RAUDOITTEIDEN KOKONAISPAINO YHTEENSÄ [kg]: 47.7

REINFORCING BAR LIST CAST UNIT CODE: CFO/16 PCS: 1

REINFORCING BARS				D	L	dL	WEIGHT	BENDING DIMENSIONS [mm]									COMMENT
TYPE	POS	PCS	GRADE	[mm]	[mm]	[mm]	SUM [kg]	a	b	c	d	e	u	v	x	TD	
A	8	8	AISI304	12	590		4.2	585									
A	17	4	AISI304	12	600		2.1	601									
D	22	16	AISI304	8	1040		6.5	500	70	500						36	
D	33	12	AISI304	16	1620		30.8	490	716	490						80	
D	76	5	AISI304	12	1830		8.1	600	670	600						32	

RAUDOITTEIDEN KOKONAISPAINO YHTEENSÄ [kg]: 51.7

REINFORCING BAR LIST CAST UNIT CODE: CFO/17 PCS: 1

REINFORCING BARS				D	L	dL	WEIGHT	BENDING DIMENSIONS [mm]									COMMENT
TYPE	POS	PCS	GRADE	[mm]	[mm]	[mm]	SUM [kg]	a	b	c	d	e	u	v	x	TD	
A	7	3	AISI304	12	610		1.6	609									
A	18	3	AISI304	12	620		1.7	621									
A	21	2	AISI304	12	330		0.6	334									
D	22	16	AISI304	8	1040		6.5	500	70	500						36	
D	37	6	AISI304	12	1330		7.1	450	470	450						32	
D	77	4	AISI304	16	1460		9.2	510	516	510						80	
D	78	4	AISI304	16	1480		9.4	520	516	520						80	

RAUDOITTEIDEN KOKONAISPAINO YHTEENSÄ [kg]: 36.1

CONCRETE INFORMATION

Concretes used	C35
Concrete cover 1	50±10mm

REINFORCING BAR LIST CAST UNIT CODE: CFO/1 PCS: 1

REINFORCING BARS				D	L	dL	WEIGHT	BENDING DIMENSIONS [mm]									COMMENT
TYPE	POS	PCS	GRADE	[mm]	[mm]	[mm]	SUM [kg]	a	b	c	d	e	u	v	x	TD	
A	4	20	AISI304	12	470		8.3	470									
A	15	10	AISI304	12	500		4.5	505									
D	26	160	AISI304	8	770		48.7	368	70	368						36	
D	42	36	AISI304	16	2280		129.8	370	1616	370						80	
D	43	4	AISI304	12	2430		8.6	450	1570	450						32	
D	44	24	AISI304	16	3180		120.7	370	2516	370						80	
A	74	4	AISI304	12	2490		8.8	2490									

RAUDOITTEIDEN KOKONAISPAINO YHTEENSÄ [kg]: 329.6

REINFORCING BAR LIST CAST UNIT CODE: CFO/10 PCS: 1

REINFORCING BARS				D	L	dL	WEIGHT	BENDING DIMENSIONS [mm]									COMMENT
TYPE	POS	PCS	GRADE	[mm]	[mm]	[mm]	SUM [kg]	a	b	c	d	e	u	v	x	TD	
A	5	10	AISI304	12	430		3.9	434									
A	13	8	AISI304	12	450		3.2	447									
D	24	64	AISI304	8	670		17.0	318	70	318						36	
D	30	28	AISI304	16	1090		48.4	326	516	326						80	
D	71	4	AISI304	12	1610		5.7	606	456	606						72	
D	72	8	AISI304	16	2480		31.4	300	1956	300						80	
A	73	4	AISI304	12	1920		6.8	1920									

RAUDOITTEIDEN KOKONAISPAINO YHTEENSÄ [kg]: 116.3

REINFORCING BAR LIST CAST UNIT CODE: CFO/11 PCS: 1

REINFORCING BARS				D	L	dL	WEIGHT	BENDING DIMENSIONS [mm]									COMMENT
TYPE	POS	PCS	GRADE	[mm]	[mm]	[mm]	SUM [kg]	a	b	c	d	e	u	v	x	TD	
A	5	9	AISI304	12	430		3.5	434									
A	13	7	AISI304	12	450		2.8	447									
D	25	64	AISI304	8	690		17.5	328	70	328						36	
D	30	26	AISI304	16	1090		45.0	326	516	326						80	
D	69	8	AISI304	16	2280		28.8	300	1751	300						80	
A	70	4	AISI304	12	1710		6.1	1715									
D	71	4	AISI304	12	1610		5.7	606	456	606						72	

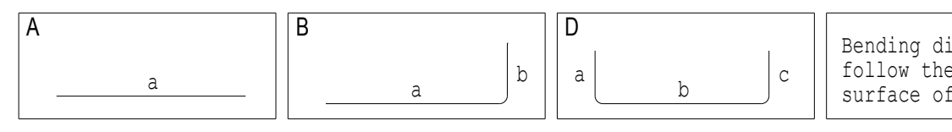
RAUDOITTEIDEN KOKONAISPAINO YHTEENSÄ [kg]: 109.3

REINFORCING BAR LIST CAST UNIT CODE: CFO/12 PCS: 1

REINFORCING BARS				D	L	dL	WEIGHT	BENDING DIMENSIONS [mm]									COMMENT
TYPE	POS	PCS	GRADE	[mm]	[mm]	[mm]	SUM [kg]	a	b	c	d	e	u	v	x	TD	
A	1	5	AISI304	12	420		1.9	418									
A	2	3	AISI304	12	430		1.1	430									
D	23	16	AISI304	8	730		4.6	348	70	348						36	
D	37	4	AISI304	12	1330		4.7	450	470	450						32	
D	50	4	AISI304	16	1090		6.9	330	516	320						80	
D	51	4	AISI304	16	1110		7.0	335	516	335						80	

RAUDOITTEIDEN KOKONAISPAINO YHTEENSÄ [kg]: 26.3

REINFORCEMENT BENDING SHAPES:



Bending dimensions follow the outer surface of bars.

MATERIAL LIST

MATERIAL	VALUE	UNIT
C35/45	8.38	m³
AISI304 ø8	258.1	kg
AISI304 ø12	264.7	kg
AISI304 ø16	1132.1	kg
HIT-HY200-A+AM-HDG8.8 S355J2	132	pcs

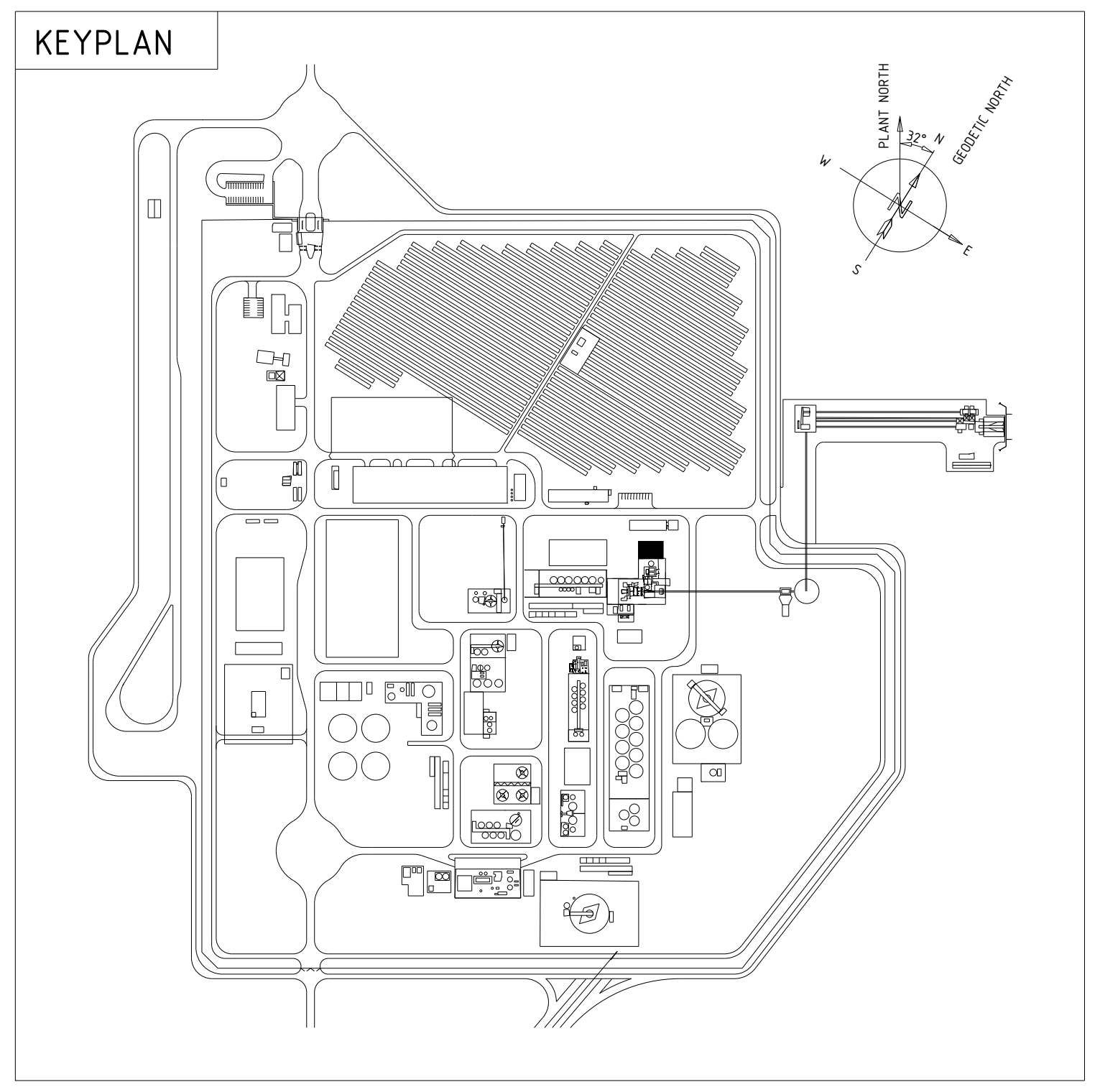
LEGEND:

R.L. - REDUCED LEVEL
T.O.C. - TOP OF CONCRETE
TL. - TOP LEVEL
B.O.F. - BOTTOM OF FOUNDATION
F.G.L. - FINISHED GROUND LEVEL
P.C.C. - PLAIN CEMENT CONCRETE
TYP - TYPICAL
MKD - MARKED
CL - CENTER LINE
BOTTOM REINF. (B)
TOP REINF. (T)
SBC - SAFE BEARING CAPACITY
MIN. - MINIMUM
REF. - REFERED
GA - GENERAL ARRANGEMENT
DWG. - DRAWING
R.C.C. - REINFORCED CEMENT CONCRETE

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS ARE IN METRES UNLESS OTHERWISE STATED
2. GRADE OF CONCRETE SHALL BE C35 (35MPa) UNLESS NOTED OTHERWISE & C15 (15MPa) FOR BLINDING CONCRETE.
3. GRADE OF REINFORCEMENT SHALL BE DEFORMED BAR ASTM A615-GRADE 60 (fy=420MPa)
4. CLEAR COVER TO MAIN REINFORCEMENT SHALL BE AS FOLLOWS:
a) FOUNDATION: 75 mm (BOTTOM & SIDE)
b) FOUNDATION: 50 mm (TOP)
c) COLUMN & PEDESTAL: 50 mm
5. MINIMUM LAP LENGTHS OF REINFORCEMENT SHALL BE 40XBAR DIAMETER
6. FOR STANDARD REINFORCEMENT DETAILS & GENERAL NOTES FOR CIVIL WORKS REFER TO DRAWING NUMBER MD-216-1000-EG-CV-DED-1005 & 1001
7. ANCHOR BOLTS AND EMBEDDED STEELWORK SHALL BE HOT DIP GALVANIZED
8. T.O.C. & TOP OF SCREED TOLERANCE SHALL BE WITHIN +/- 10MM OF DESIGN LEVEL
9. NET SAFE BEARING CAPACITY OF SOIL AT BOTTOM OF FOUNDATION IS CONSIDERED AS 375 KN/SQ.M.
10. REBARS CAN BE MOVED LOCALLY BY +/- 25mm TO AVOID INTERFERENCE WITH EMBEDDED PARTS

KEYPLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3220-EG-CV-GAD-1154	DIMENSIONAL DRAWING		
MD-216-3220-EG-CV-CAL-1162	CALCULATION REPORT		

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
D01	19.12.2023	AS-BUILT	NadSal	CemEce	NadSal	LuiPiz	
C01	22.02.2023	ISSUED FOR CONSTRUCTION	NadSal	CemEce	NadSal	LuiPiz	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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LARSEN & TOUBRO DRAWING NUMBER
REVISION

OUTOTEC DRAWING NUMBER	DOCUMENT ID	REVISION
900707_GAU01_ZG01_CTB17_32212	OU100781221	R0

DRAWN	NadSal	14.03.2023
CHECKED	CemEce	14.02.2023
ENGINEER	NadSal	14.03.2023
APPROVED	LuiPiz	14.02.2023
EMPLOYER		
SCALE	1:20 1:10	A0

MANSOURAH - MASSARAH GOLD PROJECT
REINFORCEMENT DRAWING
PIPELINE SUPPORTS FOUNDATIONS
GRINDING MILL COOLING WATER

DRAWING NUMBER MD-216-3220-EG-CV-DED-1162
REV D01

Tekla Structures